LLM Probability Concentration: How Alignment Shrinks the Generative Horizon

Chenghao Yang, Sida Li, Ari Holtzman
Data Science Institute, University of Chicago
Department of Computer Science, University of Chicago
{chenghao, listar2000, aholtzman}@uchicago.edu
Codebase Website

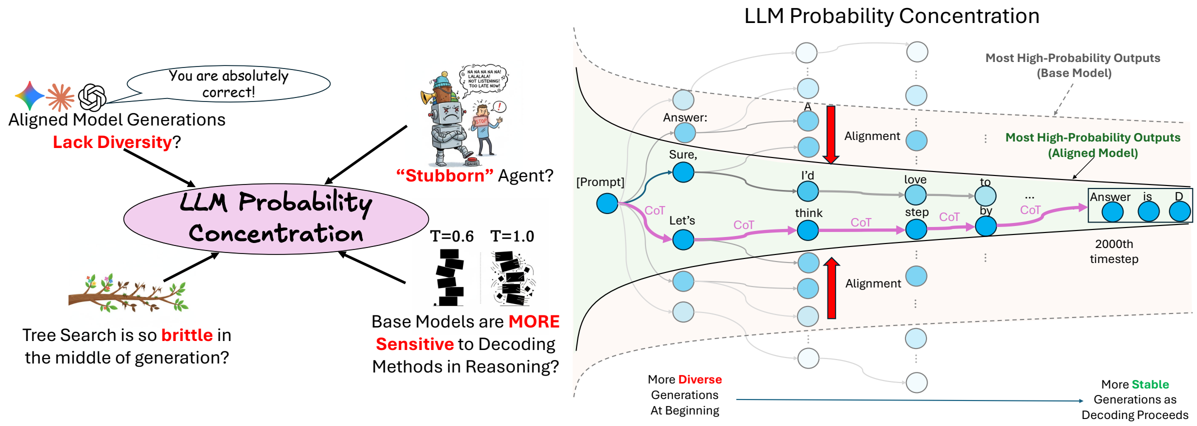
Abstract

Despite their impressive capabilities, aligned large language models (LLMs) often generate outputs that lack diversity. What drives this consistency in the generation? We investigate this phenomenon through the lens of probability concentration in the model's output distribution. To quantify this concentration, we introduce the *Branching Factor* (BF)–a token-invariant measure of the effective number of plausible next steps during generation. Our empirical analysis reveals two key findings: (1) BF often decreases as generation progresses, suggesting that LLMs become more predictable as they generate. (2) alignment tuning substantially sharpens the model's output distribution from the outset, reducing BF by a factor of 2–5 overall, and up to an order of magnitude (e.g., from 12 to 1.2) at the beginning positions. This stark reduction helps explain why aligned models often appear less sensitive to decoding strategies. Building on this insight, we find this consistency has surprising implications for complex reasoning. Aligned Chain-of-Thought (CoT) models (e.g., DeepSeek-distilled models), for instance, leverage this effect; by generating longer reasoning chains, they push generation into later, more deterministic (lower BF) stages, resulting in more stable outputs. We hypothesize that alignment tuning does not fundamentally change a model's behavior, but instead steers it toward stylistic tokens (e.g., "Sure") that unlock low-entropy trajectories already present in the base model. This view is supported by nudging experiments, which show prompting base models with such tokens can similarly reduce BF. Together, our findings establish BF as a powerful diagnostic for understanding and controlling LLM outputs - clarifying how alignment reduces variability, how CoT promotes stable generations, and how base models can be steered away from diversity.

1 Introduction

While alignment tuning improves helpfulness and safety in large language models (LLMs), it often introduces a trade-off: reduced output diversity (Padmakumar & He, 2024; Chakrabarty et al., 2024; Tian et al., 2024; Kirk et al., 2024; Lu et al., 2025) and increased determinism (Saparov & He, 2023; Song et al., 2024; Renze & Guven, 2024; Bigelow et al., 2024; West & Potts, 2025). As a result, aligned models are frequently observed to be less sensitive to different decoding strategies—a phenomenon we confirm in our own case study (§ 5.1). Similarly, Chain-of-Thought (CoT) prompting (Wei et al., 2022), while enhancing reasoning, often reduces the variance of answers. These observations point toward a common underlying phenomenon: *LLM Probability Concentration* (Figure 1a), where the model's vast potential output space collapses into a narrow set of likely trajectories.

But how should we rigorously conceptualize and measure this concentration? Autoregressive generation is inherently a traversal through a *branching tree* (Figure 1b). Standard metrics often fail to capture the global structure of this tree. Token-level entropy is too local; "model perplexity" (Jurafsky & Martin, 2025)



(a) (b)

Figure 1: (a): LLM probability concentration connects and explains several disparate yet critical phenomena in aligned LLMs. (b): A conceptual illustration of how alignment and CoT influence the generation space of LLMs. While base models begin with high output diversity, alignment tuning sharply concentrates early probability mass, leading to more stable outputs. CoT extends this effect into later positions, flattening output sample variation and reducing sensitivity to decoding.

measures fit to a reference dataset rather than the model's own generative breadth; and surface-level diversity metrics (e.g., n-gram diversity) are often confounded by vocabulary size and output length.

To address this, we operationalize a principled concept from information theory: *distribution perplexity* (or exponentiated entropy). Specifically, we define the **Branching Factor (BF)** as the *length-normalized* distribution perplexity (i.e., exponentiated entropy rate). This metric quantifies the effective number of viable next-token choices available to the model on average, providing a "microscopic" lens on the tree's expansion rate. Crucially, we do not propose BF as a novel mathematical metric; rather, we introduce it as a unifying framework to quantify LLM probability concentration. By leveraging the asymptotic closeness between length-averaged log-probability and entropy (Mudireddy et al., 2024), we can estimate BF efficiently from the model's own naturally sampled sequences, avoiding the need for teacher forcing or exhaustive enumeration.

The core contribution of this work is using the BF framework to provide a *unified explanation* for disparate LLM behaviors. We show that seemingly unconnected phenomena, reduced diversity in alignment, decoding insensitivity, and CoT stability, are all manifestations of probability concentration dynamics. Specifically:

① **Alignment Constrains the Branching Space:** We find that alignment tuning (e.g., RLHF) significantly reduces BF, typically by a factor of 2–5 overall, and up to an order of magnitude (e.g., $12 \rightarrow 1.2$) at the beginning positions, depending on alignment intensity. This reduction provides a quantitative mechanism for why aligned models are so insensitive to decoding parameters: there are simply fewer viable branches to prune.

② **Dynamic Concentration and CoT Stability:** We observe that BF typically declines over the course of generation, indicating that models "commit" to narrower trajectories as they generate. This explains the stabilizing effect of Chain-of-Thought: by encouraging long reasoning chains, CoT pushes the critical answer generation into later, lower-BF regions of the tree, resulting in more deterministic outcomes.

③ **Alignment Surfaces Latent Low-Entropy Paths:** Finally, we investigate *how* alignment achieves this concentration. Through "nudging" experiments, we show that conditioning a base model on a short, aligned-style prefix is sufficient to trigger a rapid drop in BF. This suggests that alignment does not fundamentally reshape the model's manifold but rather steers generation toward low-entropy subspaces that are already latent in the pre-trained model.

In summary, by viewing generation through the lens of BF, we move beyond reporting *that* aligned models are less diverse, to explaining *how* this concentration emerges from the underlying probabilistic structure.

2 Background

Autoregressive Language Models. LLMs are typically trained to predict the next token and the probability of output $P(y_{1:N}|x;\theta)$ can be decomposed as: $P(y_{1:N}|x;\theta) = \Pi_{t=1}^{N} P(y_t|[x, y_{1:t-1}];\theta)$, where $y_{1:t-1}$ is the output up to position $t-1$, θ is the model parameter, and x is the prompt (treated as fixed). Each output sample is generated via token-by-token sampling, and the generation of multiple samples naturally forms a search tree (Yao et al., 2023; Hao et al., 2023; Wan et al., 2024). Modern LLMs go through multiple training stages. In this paper, we would use *base models* to refer to the models trained without *alignment tuning* techniques (Touvron et al., 2023), including instruction tuning and Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022) (e.g., "Llama-2-13B" (Touvron et al., 2023)) and *aligned models* to refer to models undergoing these additional fine-tuning stages (e.g., "Llama-2-13B-Chat").

LLM Decoding and Entropy. Though LLMs are trained with a large vocabulary size $|V|$, the desired tokens often concentrate on a much smaller set of tokens under distribution $P(y_t|x, y_{1:t-1};\theta)$. Common decoding methods (Holtzman et al., 2020; Hewitt et al., 2022) utilize this observation and propose various heuristics to truncate vocabulary V as V_t at each step t. The next token is then sampled from the renormalized distribution $\tilde{P}(y_t|[x, y_{1:t-1}];\theta) = \mathbb{1}(y_t \in V_t)\frac{P(y_t|x,y_{1:t-1};\theta)}{\sum_{y_t \in V_t} P(y_t|x,y_{1:t-1};\theta)}$. Since tokens are sampled from the truncated distribution $\tilde{P}$,[1] we use $\tilde{P}$ to compute the empirical (token-level) entropy $\tilde{H}$ for a given prefix instance $y_{1:t-1}$:[2]

$$\tilde{H}(\mathrm{Y}_t|[x, y_{1:t-1}];\theta) = -\sum_{y_t} \tilde{P}(y_t|[x, y_{1:t-1}];\theta) \log \tilde{P}(y_t|[x, y_{1:t-1}];\theta) \tag{1}$$

Note that $\tilde{H}$ is a *random variable* that depends on the specific realization of the prefix sequence $Y_{1:t-1} = y_{1:t-1}$. The more common notion of *conditional entropy* is thus the expectation of $\tilde{H}$ over all possible $y_{1:t-1}$:

$$\tilde{H}(\mathrm{Y}_t|[x, \mathrm{Y}_{1:t-1}];\theta) = \mathbb{E}_{y_{1:t-1}} \tilde{H}(\mathrm{Y}_t|[x, y_{1:t-1}];\theta) \tag{2}$$

Conventionally, we use uppercase Y to denote the *random variable* for an output and lowercase y for its specific realization. Finally, along a realized sequence $y_{1:t-1}$, we define the *realized entropy* as[3]

$$h_{\text{realized}}(y_{1:N}) \stackrel{\text{def}}{=} \sum_{t=1}^{N} \tilde{H}(\mathrm{Y}_t|y_{1:t-1};\theta). \tag{3}$$

This metric and its sequence-level reductions (e.g., mean pooling) are of profound practical importance: while estimating the full conditional entropy $\tilde{H}(\mathrm{Y}_t|x;\mathrm{Y}_{<t})$ requires marginalizing over an exponential space of prefix trajectories, h_{realized} serves as the widely-adopted tractable proxy for quantifying generation uncertainty (Kuhn et al., 2023; Farquhar et al., 2024), generation diversity and creativity (Dušek et al., 2020; West & Potts, 2025) and controlling exploration in RLVR (Cheng et al., 2025; Cui et al., 2025; Wang et al., 2025).[4] Linearity of expectation and the chain rule for entropy then gives us

$$\mathbb{E}_{y_{1:N}}\left[h_{\text{realized}}(y_{1:N})\right] = \sum_{t=1}^{N} \tilde{H}(\mathrm{Y}_t|[x, \mathrm{Y}_{1:t-1}];\theta) = \tilde{H}(Y_{1:N}|x;\theta), \tag{4}$$

i.e. h_{realized} is an unbiased estimator of the marginal entropy of the whole generative process.

A Note on Practical Text Generation. Throughout this paper, we model an LLM as generating sequences of a fixed maximum length N, with realizations denoted by $y_{1:N}$, and some theoretical results consider the asymptotic regime $N \to \infty$. In practice, however, generation often terminates early, producing a shorter sequence $y_{1:n}$ with $n < N$. Our framework accommodates this by defining an *equivalent sequence* $\hat{y}_{1:N}$ such that $\hat{y}_t = y_t$ for $t \leq n$, and $\hat{y}_{n+1:N}$ consists of repeated special tokens (e.g., EOS) indicating termination. We further assume $\tilde{P}\big(Y_{n+1:N} = \hat{y}_{n+1:N} \mid [x, y_{1:n}];\theta\big) = 1$, which can be viewed as a property of pretrained LLMs. Under this convention, the probability of the equivalent sequence satisfies $\tilde{P}(\hat{y}_{1:N} \mid x;\theta) = \tilde{P}(y_{1:n} \mid x;\theta)$. Consequently, without loss of generality, we may treat all generations as having length N.

[1] Our main experiments employ mild decoding settings (T=1.0, p=0.9). These settings approximate the full distribution, align with standard evaluation practices, and ensure coherent generation from base models. Stronger truncation settings are explicitly noted where applied.

[2] The common convention setting $0 \log 0 = 0$ for entropy computation is followed.

[3] For notational brevity, we hereafter omit explicit conditioning on the input x when the context is clear.

[4] See more related work discussions in § 7